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Node Embeddings...

→ Representation of Node into a vector.

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#

Cora dataset

Contains 2708 nodes, 10556 ed, & 1433 fea

→ first MLP implemented
[Multi-layer perception.] → this is without GNN. (or) Graph info
achived 44% accuracy.

Observed that raw node feature alone are not sufficient for accurate classification.

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Now GNN [Graph Neural Network]

- GCN
- GAT
- GIN

→ GCN :- Graph Convalutional Network

aggregates info from neighbour-ing nodes for better learning.

→ GAT :- Graph Attention

uses an attention mechanism to ~~[illegible]~~ priotize imp neighbour

→ GIN :- Graph

provides stronger structural representation learning

Next Generated the Node Embeddings using graph-based learning techniques.

(*) GNN Embeddings showed better Class seperation & clustering the raw features.

(*) I compared traditional ML with Graph-based approaches. So Conclusion is that combination of node features & graph structure had significantly improved node classification performance

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MLP :-

- Uses only node features
- ~44% accuracy
- Fastest runtime.
- Low CC [computational cost].
- Can't capture graph relationships

GCN :-

- Aggregates neighbour info
  - High (75)
- Moderate (Runtime)
- Moderate [CC].
- Best balance for accuracy & efficiency

GAT :-

- Uses attention mechanism
- Higher than GCN
- Slow [Runtime]
- High [CC]

- Learns imp neighbour contribution

Deep & Wired

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Hyperparameter.

GIN:-

- Strong structural representation learning
- Best | competive.
- Highest in both Runtime & CC
- Produced strongest embeddings & Cluster seperation.

Real-time Scenario's...

low-resource Sys: use MLP
because its very fast & lightweight but lower accuracy.

Large-scale practical application: use GCN

Best balance b/w accuracy, runtime & Computational cost.

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* Applications like needed interpretability

GAT Performs good because Attention mechanism helps identify imp neighbour.

* Complex graph pattern analysis

GIN because this is good in strongest structural learning & representation power.

Model	Real-time Suitability
MLP	Fast but less intelligent
GCN	Most paractical overall
GAT	Accurate but compu expensive
GIN	powerfull but heavy for deployment.

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For Real-world deployment, GCN [Graph Convolutional Network] is generally prefered because it achives good accuracy with lower compunational overhead compared to GAT & GIN.

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Graph Sage :- ...

new thing

This improves scalability in graph neural network by sampling & aggragating neighbour info instead of processing the full graph, making it highly suitable for high-scale (or) large scale & real-time appli..

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Real-time deployment, large-scale prediction

$\underbrace{\hphantom{Real-time deployment, large-scale prediction}}$

* GCN

Highest accuracy research tasks, Attention based analysis

$\underbrace{\hphantom{Highest accuracy research tasks, Attention based analysis}}$

* GAT